Low-Crosstalk Silicon Photonic Optical Phased Array for 1550 nm FMCW LiDAR: Project Proposal

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I. Introduction

LiDAR (light detection and ranging) is a remote-sensing technique that measures the distance to objects by sending out laser light and analyzing the light reflected back. Scanning the beam across a scene builds a 3-D map, which is why LiDAR is used in autonomous vehicles, drones and robotics. Frequency-modulated continuous-wave (FMCW) LiDAR sweeps the laser frequency and compares the returning light with the outgoing light. From that comparison it obtains both distance and velocity, and it is largely immune to sunlight and to interference from other LiDARs [1]. To scan the beam without moving parts, an optical phased array (OPA) places a row of tiny light emitters on a silicon chip and gives each one an adjustable phase. Tilting the phase across the row steers the beam, much as a phased-array radar does [2–4].

The main difficulty is emitter spacing. To steer across $\pm 45^\circ$ at 1550 nm without unwanted duplicate beams (*grating lobes*), a uniformly spaced array needs emitters less than about $0.9\,\mu$m apart. At that spacing, light leaks between neighboring waveguides (optical crosstalk) and heat leaks between neighboring phase shifters (thermal crosstalk), and both distort the beam. At a safer spacing of $1.5\,\mu$m, a uniform array can steer only about $\pm 2^\circ$ before grating lobes appear.

This project addresses that trade-off. We will design a 1×32 OPA transmitter whose emitters stay at least $1.5\,\mu$m apart but are *irregularly* spaced to suppress grating lobes. We will verify the design in simulation, from individual components up to the full circuit, and deliver a foundry-compatible mask layout.

II. Related Work

Uniform silicon OPAs. The first silicon OPA used 16 emitters at $2\,\mu$m spacing and steered only 2.3° [5]. That small range is exactly the spacing problem described above. Later work scaled the approach to 4096 emitters [2] and demonstrated FMCW ranging [3]. These arrays are compact and manufacturable in CMOS foundries, but they still trade steering range against crosstalk, and each heater-based phase shifter draws about 25 mW [6].

Irregular (aperiodic) arrays. Spacing emitters irregularly spreads the energy of the grating lobes into a low background of side lobes. This allows wide spacing with no strong duplicate beams, and the background falls as the number of emitters grows [7]. A 128-emitter irregular array demonstrated wide, high-resolution steering without grating lobes [8]. The drawbacks are that side-lobe suppression is limited by the emitter count and that it degrades if positions or phases are poorly chosen.

Silicon nitride platforms. Silicon nitride handles much higher optical power than silicon [9], because silicon suffers two-photon absorption at high intensity [10]. Nitride, however, needs larger bends and wider spacing, and its heaters are less efficient. Hybrid designs use nitride only to distribute power and silicon for everything else [11], at the cost of an extra layer.

Crosstalk suppression. Varying the width of neighboring waveguides ("superlattices") reduces their coupling enough to allow very tight spacing [12]. Engineered claddings made of sub-wavelength silicon strips achieve the same by shortening the evanescent field [13]. Both work well, but they need either several waveguide designs or very small features. Reviews list crosstalk, phase errors and power handling as the main open problems [4]. We focus on crosstalk, using the irregular-array approach because it works with standard waveguides and design rules.

III. Project Objectives

We will model, simulate and lay out a 1×32 OPA transmitter on a standard 220-nm silicon-on-insulator (SOI) platform at 1550 nm, with 1×16 as a fallback. Fabrication is outside the scope of this semester.

O1 – Array design (Jordan). Choose 32 irregular emitter positions, at least $1.5\,\mu$m apart, that keep every side lobe at least 10 dB below the main beam across $\pm 45^\circ$ of steering.

O2 – Component design (Jacob). Design the emitter (a grating antenna), a 1-to-2 power splitter and a heater-based phase shifter. Targets are optical crosstalk between neighbors below -20 dB (under 1% of the power), splitter loss below 0.3 dB, and a phase shifter that needs at most 25 mW for a half-wave (π) shift and affects its neighbor by less than 5%.

O3 – Full-circuit simulation (Jacob). Connect the components into the complete 1×32 circuit and show that the beam steers smoothly across $\pm 45^\circ$. Include realistic manufacturing variation and show that a simple calibration recovers the beam.

O4 – Mask layout (Jordan). Produce a layout file (GDSII) that passes the design-rule check (DRC) of an open foundry process design kit (SiEPIC EBeam PDK).

O5 – LiDAR performance estimate (Jordan). Translate the simulated losses and beam quality into range resolution and maximum range.

Stretch goal (both). If time allows, estimate how much optical power the silicon splitter tree can handle before two-photon absorption becomes a problem, and compare it with a silicon nitride splitter tree.

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IV. Methodology

We will use Synopsys/Lumerical tools for device and circuit physics, Python for array modeling and for automating simulation sweeps, and KLayout for layout. All scripts share one settings file (wavelength, number of emitters, minimum spacing, PDK layers), so both members always work from the same design values.

O1 – Array design. The beam pattern of an array follows directly from the emitter positions and phases (the *array factor*). We will compute it in Python and use an optimizer to search for positions that minimize the worst side lobe over the full steering range. We will first assume ideal emitters, then repeat the optimization using the real emitter pattern from O2.

O2 – Components. For crosstalk, we will first estimate coupling between neighboring waveguides with methods from class (effective index method and Marcatili's approximation). We will then confirm the estimates with Lumerical MODE over a range of spacings and widths. The quick estimate and the full simulation check each other. The grating antenna will be tuned in Lumerical FDTD (period, etch depth, fill factor) for efficient upward emission, and FDTD will also give its emission pattern for O1. The splitter (a multimode interference coupler) will be designed in Lumerical MODE/EME. Heater power and thermal crosstalk will be simulated in Lumerical HEAT to choose how far apart the phase shifters must be. Selected results will be cross-checked with Synopsys RSoft or the open-source solver MEEP.

O3 – Full circuit. The simulated component responses (S-parameters) will be imported into Lumerical INTERCONNECT and connected into the full 1×32 circuit. For each steering angle, Python will set the heater powers, read the light leaving each emitter and compute the resulting beam. We will then add random width and thickness variations to mimic fabrication, and test a calibration routine that adjusts the heaters to restore the beam, as would be done on a real chip.

O4 – Layout. We will build the circuit in KLayout (with the gdsfactory Python library), placing emitters at the O1 positions and heaters at the O2 spacing. We will run the PDK's DRC and fix all violations, then confirm that the layout's connectivity matches the O3 circuit.

O5 – LiDAR estimate. Using the circuit loss and beam quality from O1–O3, we will estimate range resolution and maximum range with standard FMCW LiDAR relations [1].

Risks. If full 3-D simulations are too slow, we will simulate short sections and use faster 2.5-D solvers. If a target is not met, we will report the trade-off we actually achieve (for example, side-lobe level versus spacing) rather than change scope late in the semester.

V. Collaboration Plan

Jacob leads the Lumerical component and circuit work (O2, O3). **Jordan** leads array design, layout and the LiDAR estimate (O1, O4, O5). The two halves meet at three hand-offs: Jacob provides the emitter pattern and component models, and Jordan provides the emitter positions. Both share the stretch goal, the presentations, the final IEEE paper and the weekly status checks. Table 1 gives each person's weekly tasks.

VI. Timeline

Table 1 assigns one deliverable to each weekly status check. Technical work finishes on Nov. 15 and the week of Nov. 23 is kept free as buffer, so a one-week slip still meets the Nov. 30 deadline.

Table 1: Weekly Timeline (Sep. 25 – Nov. 30, 2026)

Wk	Dates	Jacob	Jordan	Status-check deliverable
1	Sep 25–Oct 4	Set up Lumerical; class-method crosstalk estimates	Set up Python, KLayout and PDK; array model	Tools working
2	Oct 5–11	MODE crosstalk vs. spacing	First emitter positions	Crosstalk-vs-spacing results; slides
3	Oct 12–18	Grating antenna (2-D FDTD)	Basic layout cells	**Proposal presentation**
4	Oct 19–25	Antenna emission pattern (3-D FDTD)	Re-optimize positions with real pattern	Updated beam pattern
5	Oct 26–Nov 1	Splitter (EME); heater (HEAT)	Waveguide routing; LiDAR estimate	**M1:** components finished
6	Nov 2–8	Full 1×32 circuit (INTERCONNECT)	Beam calculation from circuit; layout v1 and DRC	**M2:** beam-steering demo
7	Nov 9–15	Fabrication variation; calibration	DRC-clean layout; connectivity check	**M3:** technical work complete
8	Nov 16–22	Paper (O2, O3); stretch goal	Paper (O1, O4, O5); stretch goal	**M4:** full paper draft
9	Nov 23–29	Buffer for delays; revisions; final slides (Thanksgiving Nov 26)		Final draft reviewed
–	Nov 30	Submit		**M5:** paper and layout

REFERENCES

[1] B. Behroozpour, P. A. M. Sandborn, M. C. Wu, and B. E. Boser, "Lidar system architectures and circuits," *IEEE Commun. Mag.*, vol. 55, no. 10, pp. 135–142, 2017.

[2] J. Sun, E. Timurdogan, A. Yaacobi, E. S. Hosseini, and M. R. Watts, "Large-scale nanophotonic phased array," *Nature*, vol. 493, no. 7431, pp. 195–199, 2013.

[3] C. V. Poulton, A. Yaacobi, D. B. Cole, M. J. Byrd, M. Raval, D. Vermeulen, and M. R. Watts, "Coherent solid-state LIDAR with silicon photonic optical phased arrays," *Opt. Lett.*, vol. 42, no. 20, pp. 4091–4094, 2017.

[4] M. Hu, Y. Pang, and L. Gao, "Advances in silicon-based integrated lidar," *Sensors*, vol. 23, no. 13, p. 5920, 2023.

[5] K. Van Acoleyen, W. Bogaerts, J. Jágerská, N. Le Thomas, R. Houdré, and R. Baets, "Off-chip beam steering with a one-dimensional optical phased array on silicon-on-insulator," *Opt. Lett.*, vol. 34, no. 9, pp. 1477–1479, 2009.

[6] N. C. Harris, Y. Ma, J. Mower, T. Baehr-Jones, D. Englund, M. Hochberg, and C. Galland, "Efficient, compact and low loss thermo-optic phase shifter in silicon," *Opt. Express*, vol. 22, no. 9, pp. 10487–10493, 2014.

[7] T. Komljenovic, R. Helkey, L. Coldren, and J. E. Bowers, "Sparse aperiodic arrays for optical beam forming and LIDAR," *Opt. Express*, vol. 25, no. 3, pp. 2511–2528, 2017.

[8] D. N. Hutchison, J. Sun, J. K. Doylend, R. Kumar, J. Heck, W. Kim, C. T. Phare, A. Feshali, and H. Rong, "High-resolution aliasing-free optical beam steering," *Optica*, vol. 3, no. 8, pp. 887–890, 2016.

[9] C. V. Poulton, M. J. Byrd, M. Raval, Z. Su, N. Li, E. Timurdogan, D. Coolbaugh, D. Vermeulen, and M. R. Watts, "Large-scale silicon nitride nanophotonic phased arrays at infrared and visible wavelengths," *Opt. Lett.*, vol. 42, no. 1, pp. 21–24, 2017.

[10] H. K. Tsang and Y. Liu, "Nonlinear optical properties of silicon waveguides," *Semicond. Sci. Technol.*, vol. 23, no. 6, p. 064007, 2008.

[11] Q. Wang, S. Wang, L. Jia, Y. Cai, W. Yue, and M. Yu, "Silicon nitride assisted 1×64 optical phased array based on a SOI platform," *Opt. Express*, vol. 29, no. 7, pp. 10509–10517, 2021.

[12] W. Song, R. Gatdula, S. Abbaslou, M. Lu, A. Stein, W. Y.-C. Lai, J. Provine, R. F. W. Pease, D. N. Christodoulides, and W. Jiang, "High-density waveguide superlattices with low crosstalk," *Nat. Commun.*, vol. 6, p. 7027, 2015.

[13] S. Jahani, S. Kim, J. Atkinson, J. C. Wirth, F. Kalhor, A. A. Noman, W. D. Newman, P. Shekhar, K. Han, V. Van, R. G. DeCorby, L. Chrostowski, M. Qi, and Z. Jacob, "Controlling evanescent waves using silicon photonic all-dielectric metamaterials for dense integration," *Nat. Commun.*, vol. 9, p. 1893, 2018.